PROJECT REPORT

Automated Network Request Management

ServiceNow Scoped Application and Network Request Portal

Date: November 2026

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Project Documentation

Prepared based on the provided project implementation and reference report structure.
Technology: ServiceNow
Application: Automated Network Request Management
Project: Automated Network Request Management

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1. INTRODUCTION

1.1 Project Overview

The Employee Raise Issue project is a ServiceNow-based scoped application designed to provide employees with a structured, user-friendly mechanism for submitting workplace IT issues or requests. The solution combines a custom Employee Raise Issue table, Service Catalog item/Record Producer, dependent choice fields, UI Policies, a dedicated Service Portal, and custom widgets.

The implementation provides a controlled path from employee interaction through the portal to creation of a backend issue record. It reduces unstructured issue reporting, improves data consistency, and provides a foundation for efficient issue assignment, tracking, and resolution.

1.2 Purpose

- Provide a standardized mechanism for employees to raise IT-related issues.
- Capture issue information in a structured backend table.
- Guide employees through relevant Category and Subcategory selections.
- Enforce field behavior and mandatory information using UI Policies.
- Provide a dedicated portal experience with custom interactive widgets.
- Connect the portal submission experience with the Employee Raise Issue backend record.

2. IDEATION PHASE

2.1 Problem Statement

Employees may need to report network, hardware, software, access, or related support issues. Without a standardized intake mechanism, issue information can be incomplete or inconsistent, making categorization, assignment, and follow-up more difficult. The project addresses this problem by providing a guided ServiceNow portal-based issue submission process.

2.2 Empathy Map Canvas

Perspective	Project Interpretation
Says	I need a simple way to report an issue and provide the necessary details.
Thinks	I want to know that my request was submitted correctly and routed appropriately.
Does	Logs into the employee portal, selects an issue category, enters a description, and submits the request.
Feels	Expects a quick, clear, and reliable support experience.
Pain Points	Unstructured requests, irrelevant choices, missing descriptions, and unclear submission flow.

Perspective	Project Interpretation
Needs	A guided form, controlled choices, clear navigation, and consistent record creation.

2.3 Brainstorming

- Create a scoped ServiceNow application for employee issue management.
- Create a custom table with structured issue fields.
- Use a Service Catalog item and Record Producer for guided submission.
- Use dependent Category/Subcategory choices to prevent irrelevant selections.
- Use UI Policies to make required fields mandatory under appropriate conditions.
- Provide a dedicated Service Portal and customized Employee Center homepage.
- Use custom widgets for a visually engaging portal and direct issue-raising navigation.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Employee Action	System Response	Expected Outcome
Access	Open Requesting Portal	Loads Employee Center homepage	Employee can access services
Discover	Scroll/search portal	Displays custom widgets and navigation	Employee finds Raise an Issue
Raise	Click Raise Ticket	Redirects to Record Producer	Issue form opens
Enter	Provide requester, category, subcategory and description	Applies field rules and dependencies	Valid data is captured
Submit	Click Submit	Creates Employee Raise Issue record	Issue is registered
Follow-up	Issue is processed	State/priority/assignment information supports handling	Issue can be managed to resolution

3.2 Solution Requirements

Functional Requirements

FR No.	Functional Requirement	Sub-Requirement / Expected Behavior
FR-1	Application setup	Create scoped application named Employee Center.
FR-2	Issue record creation	Create Employee Raise Issue custom table with auto-number.
FR-3	Issue data capture	Capture requester, category, subcategory, short description,

FR No.	Functional Requirement	Sub-Requirement / Expected Behavior
		state, priority and assignment group.
FR-4	Controlled choices	Display relevant subcategories based on selected category.
FR-5	Dynamic field validation	Make Short Description mandatory when State is New.
FR-6	Record integrity	Keep Number field read-only.
FR-7	Catalog submission	Provide Raise Employee Issue catalog item with mapped variables.
FR-8	Portal access	Provide Requesting Portal and Employee Center homepage.
FR-9	Widget navigation	Allow Raise Ticket widget to redirect to the Record Producer.
FR-10	Testing	Validate portal navigation, submission and configured field restrictions.

Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	Simple guided portal experience for employees.
NFR-2	Data Consistency	Controlled choices and mapped variables reduce inconsistent entries.
NFR-3	Reliability	Configuration should consistently create the intended backend record.
NFR-4	Maintainability	ServiceNow configuration is separated into tables, policies, catalog items and widgets.
NFR-5	Security	Use ServiceNow platform access controls and scoped application boundaries.
NFR-6	Performance	Portal widgets and navigation should provide a responsive user experience.

3.3 Data Flow Diagram

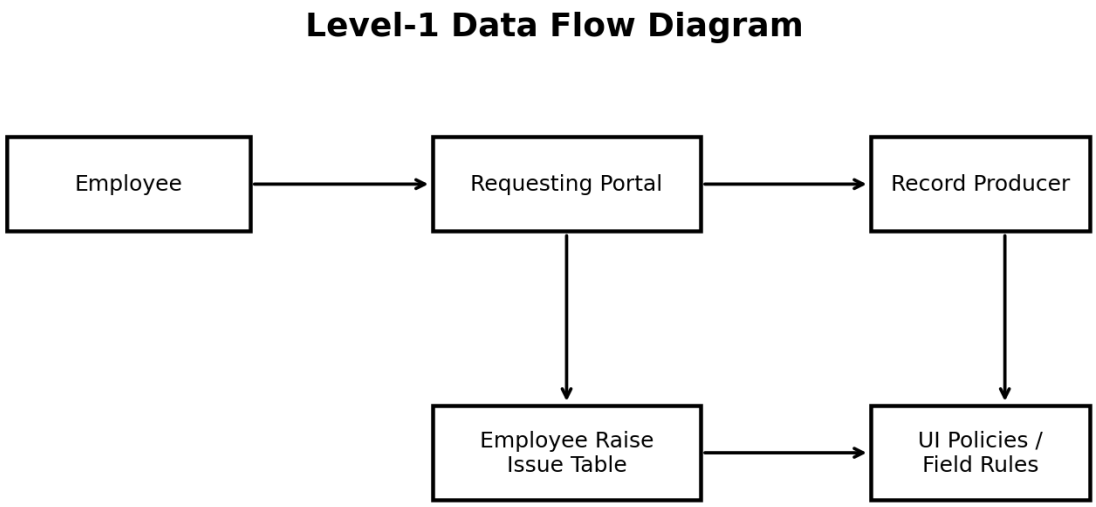


Figure 1: High-level flow from employee portal submission to backend issue record.

3.4 Technology Stack

Component	Technology / Configuration	Purpose
Platform	ServiceNow	Application platform and service management environment.
Application	Employee Center scoped application	Container for the custom solution.
Data Layer	Employee Raise Issue custom table	Stores submitted issue information.
Service Catalog	Raise Employee Issue	Provides guided issue submission.
Record Producer	Raise Employee Issue flow	Creates/submits the backend issue record from the portal experience.
Portal	Service Portal / Requesting Portal	Employee-facing interface.
Portal Design	Employee Center Homepage	Existing page customized with widgets.
Widget UI	HTML, CSS	Custom portal presentation and animation.
Client Logic	Service Portal Client Script / AngularJS	Redirects the user to the configured Record Producer.
Configuration	UI Policies / Dictionary dependency	Controls field behavior and dependent choices.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

Parameter	Project Fit
Problem	Employees need a consistent and simple way to submit workplace IT issues.
Solution	A ServiceNow Employee Raise Issue application with portal-based submission.
Customer Benefit	Guided issue entry, controlled choices and clearer submission flow.
Operational Benefit	Structured backend records support categorization and issue handling.
Uniqueness	Combines ServiceNow configuration with a dedicated portal and custom interactive widgets.
Scalability	Additional issue fields, categories, assignment logic, notifications and reporting can be added later.

4.2 Proposed Solution

The proposed solution is a scoped ServiceNow application named Employee Center. A custom Employee Raise Issue table stores issue records. A Raise Employee Issue Service Catalog item provides variables mapped to the table. A Requesting Portal and customized Employee Center homepage expose the functionality to employees. UI Policies and dependent choices enforce appropriate form behavior, while custom widgets improve navigation and presentation.

4.3 Solution Architecture

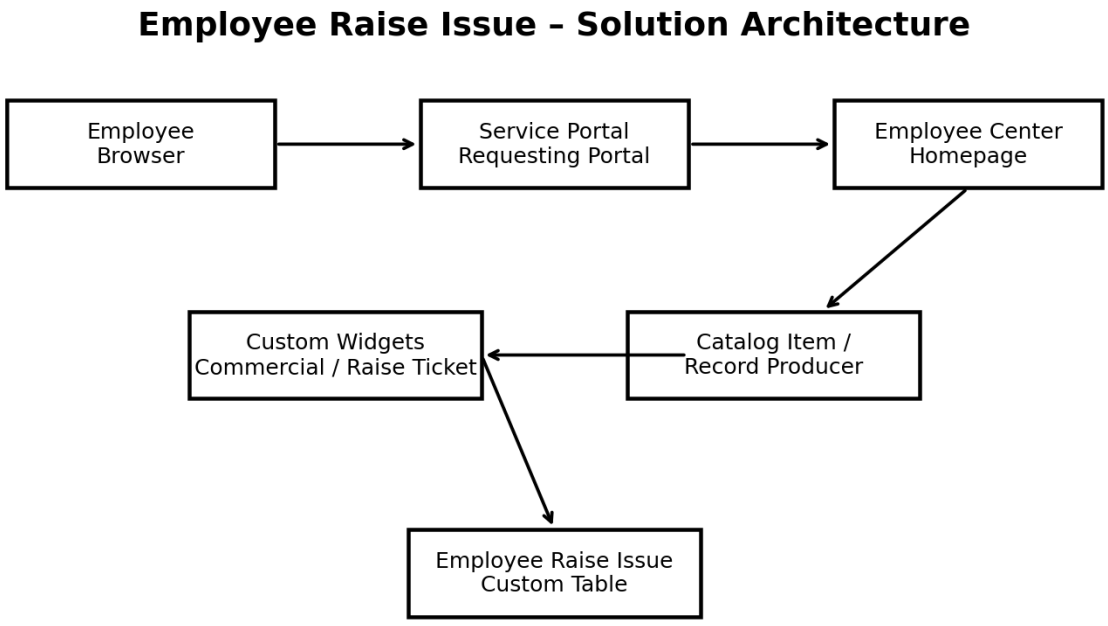


Figure 2: Employee Raise Issue solution architecture.

Core Data Model

Field	Type	Configuration / Values
Number	Auto Number	Automatically generated issue identifier.
Requester	Reference	sys_user table.
Category	Choice	Network, Hardware, Software, Access.
Subcategory	Choice	VPN, Laptop, Server, Forgot password; dependent on Category.
Short Description	String	Issue summary; mandatory when State is New.
State	Choice	New, In Progress, Resolved, Closed.
Priority	Choice	Critical, High, Medium, Low.
Assignment Group	Reference	sys_user_group table.

Category / Subcategory Dependency

Category	Subcategory
Network	VPN
Hardware	Laptop
Software	Server
Access	Forgot password

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The implementation can be organized into short iterative work packages so that the core application is established first and the portal experience is added afterward.

Phase / Sprint	Major Tasks	Deliverable
Sprint 1 – Foundation	Create scoped application; create Employee Raise Issue table; configure fields and auto-number.	Working application and data model.
Sprint 2 – Rules	Configure UI Policies; configure Category/Subcategory dependency; validate field behavior.	Validated form rules and controlled choices.
Sprint 3 – Service Catalog	Create Raise Employee Issue catalog item; create and map variables; configure Record Producer navigation.	Guided issue submission.
Sprint 4 – Portal	Create Requesting Portal; customize Employee Center homepage; create Commercial and Link Redirect widgets.	Employee-facing portal experience.
Sprint 5 – Testing	Test navigation, widget behavior, Record Producer, field restrictions and submission.	Validated end-to-end solution.

Project Tracker

Work Item	Status	Validation
Scoped application	Completed	Employee Center created and selected as current application.
Custom table	Completed	Employee Raise Issue table created.
Fields	Completed	Required fields configured.
UI Policies	Completed	Short Description mandatory for New; Number read-only.
Dependent choices	Completed	Category controls Subcategory.
Catalog item	Completed	Raise Employee Issue created with mapped variables.
Portal	Completed	Requesting Portal configured.
Custom widgets	Completed	Commercial and Link Redirect widgets created.
Testing	Completed	Portal navigation and issue submission validated.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing focused on functional correctness and user interaction rather than load benchmarking. The implemented portal was exercised through the configured Service Portal URL, and the issue-raising flow was validated from widget interaction through Record Producer submission.

Test ID	Test Scenario	Expected Result	Status
TC-01	Open Requesting Portal	Portal loads and configured branding/images are visible.	Pass
TC-02	View homepage widgets	Commercial and Raise Ticket widgets appear in configured positions.	Pass
TC-03	Hover over widgets	Floating/hover animations are visible.	Pass
TC-04	Click Raise Ticket	User is redirected to the configured Record Producer.	Pass
TC-05	Category selection	Category choices are displayed correctly.	Pass
TC-06	Dependent Subcategory	Only relevant Subcategory is available for the selected Category.	Pass
TC-07	State = New	Short Description becomes mandatory.	Pass
TC-08	Number field	Number remains read-only.	Pass

Test ID	Test Scenario	Expected Result	Status
TC-09	Submit issue	Issue details are submitted and backend record is created.	Pass
TC-10	End-to-end flow	Portal-to-record process completes without functional interruption.	Pass

Performance note: the project was validated for normal portal interaction and response behavior. Formal concurrent-user/load testing was not part of the supplied implementation evidence.

7. RESULTS

7.1 Output Screenshots

The reference implementation includes portal and widget testing. Actual screenshots were not included with the supplied project description, so the following locations are reserved for project screenshots.

Figure	Recommended Screenshot
Figure 3	Requesting Portal homepage showing configured logo/image and navigation.
Figure 4	Employee Center homepage with Commercial Widget.
Figure 5	Raise Ticket / Link Redirect widget with hover effect.
Figure 6	Raise Employee Issue Record Producer.
Figure 7	Category and dependent Subcategory fields.
Figure 8	State = New with Short Description shown as mandatory.
Figure 9	Number field shown as read-only.
Figure 10	Submitted Employee Raise Issue backend record.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Standardizes employee issue submission.
- Improves data quality through structured fields and dependent choices.
- Reduces irrelevant Category/Subcategory combinations.
- Automatically generates issue numbers.
- Enforces mandatory and read-only field behavior using UI Policies.
- Provides a dedicated employee-facing portal.
- Offers direct navigation from a custom widget to the issue Record Producer.
- Provides a scalable ServiceNow foundation for future workflow and reporting enhancements.

Disadvantages / Current Limitations

- The supplied implementation does not document advanced assignment automation or notification workflows.
- Formal load/performance benchmarking with concurrent users was not provided.
- Portal screenshots and production deployment evidence were not supplied.
- Additional controls may be required before production use, depending on organizational security and access requirements.

9. CONCLUSION

By implementing a custom scoped application, custom table, UI Policies, dependent choices, Service Catalog/Record Producer, and a dedicated Service Portal, the Employee Raise Issue solution provides a structured mechanism for employees to report workplace IT issues.

The solution standardizes issue submission, enforces business rules through UI Policies, controls category and subcategory selection, and connects the employee-facing portal with backend ServiceNow records. End-to-end validation confirmed the intended navigation, widget behavior, field restrictions, and issue submission flow.

Overall, the implementation reduces manual effort and data inconsistencies while improving the employee experience for raising and managing support issues.

10. FUTURE SCOPE

- Automated assignment of issues to groups based on Category, Subcategory or Priority.
- Email and ServiceNow notifications for submission, assignment, status changes and resolution.
- Approval or escalation workflows for selected issue types.
- Employee-facing issue tracking and status history.
- Reports and dashboards for issue volume, priority, category and resolution time.
- Attachment support for screenshots and supporting evidence.
- Additional dependent categories and subcategories.
- Role-based access and more granular security controls.
- Knowledge article recommendations before an employee submits a ticket.
- Formal performance testing and monitoring under concurrent user load.

11. APPENDIX

A. Application Creation

1. Navigate to All menus → App Engine → ServiceNow Studio.
2. Select Create → App.
3. Set Name to Employee Center and Scope to Scoped.
4. Continue, return to the home screen, reload, and confirm Employee Center is the current application.

B. Table and Field Configuration

5. Navigate to Tables under System Definition and select New.
6. Set Label to Employee raises issue and enable Auto number.
7. Open the form and use Configure → Form Design to add the required fields.
8. Configure reference fields to sys_user and sys_user_group where applicable.
9. Configure Category and Subcategory as choice fields and define the required choices.

C. UI Policies

Policy	Condition	Action
Short Description Mandatory	State is New	Short description Mandatory = True.
Number Read Only	No filter condition	Number Read only = True.

D. Service Catalog Item and Variables

Variable	Type	Mapped Field
Requester	Reference	Requester / sys_user
Category	Choice	Category
Subcategory	Choice	Subcategory
State	Choice	State
Short description	Single Line Text	Short description

E. Custom Widget – Commercial Widget

The widget uses HTML and CSS to display a service-information card with gradient styling, hover movement, floating text, animated glow, and staggered list-item animations.

```
<div class="commercial-widget">
  <h2>Our Commercial Services</h2>
  <p>
    We provide end-to-end IT services, support, and issue resolution
    through a streamlined Service Portal experience.
  </p>
  <ul>
    <li>✓ 24/7 Customer Support</li>
    <li>✓ Faster Issue Resolution</li>
    <li>✓ Transparent Tracking</li>
    <li>✓ Secure & Reliable</li>
  </ul>
</div>
```

F. Custom Widget – Link Redirect / Raise Ticket

The widget provides an interactive Raise an Issue card. Its Client Script redirects the user to the configured Record Producer using its sys_id.

```
function ($scope, $location) {
  var c = this;
  var RECORD_PRODUCER_SYS_ID = 'ae2cd61183433210de6fc590ceaad3a4';

  c.openRecordProducer = function () {
    $location.url(
      'sp?id=sc_cat_item&sys_id=' + RECORD_PRODUCER_SYS_ID
    );
  };
}
```

G. Service Portal Configuration

Property	Configured Value
Title	Requesting Portal
URL suffix	employee_request
Homepage	ec_standard_home
Knowledge Base home page	esc_knowledge_home
Login page	login
Main menu	Employee Center Menu
404 page	404
Theme	Employee Center Coral Theme
Logo	Uploaded project image

H. Project Outcome

The final implementation provides standardized issue submission, controlled field behavior, dependent selections, portal-based navigation, custom widgets, and a connected ServiceNow backend record structure.